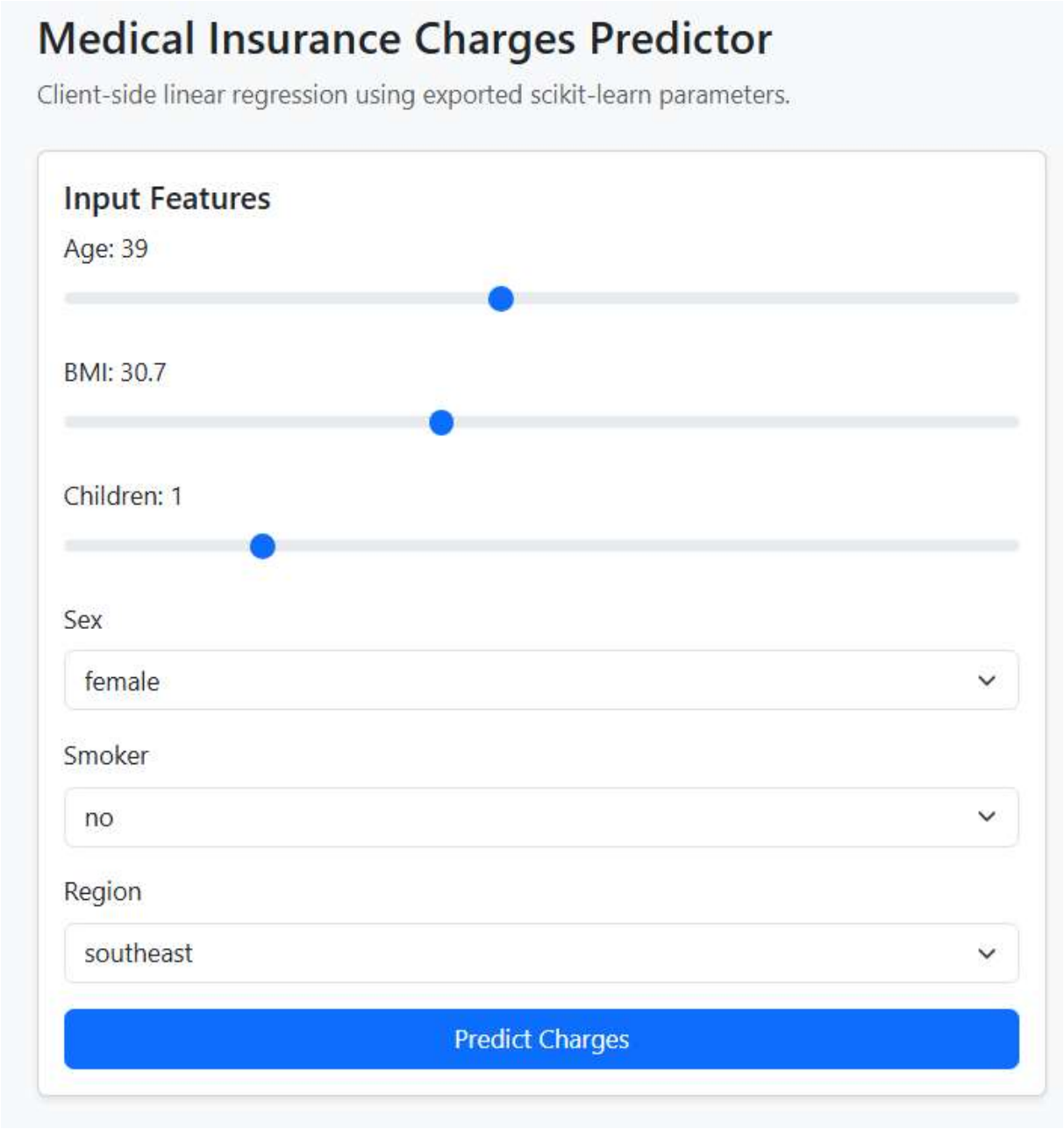
Documentation

Input Section



Prediction Output

Prediction

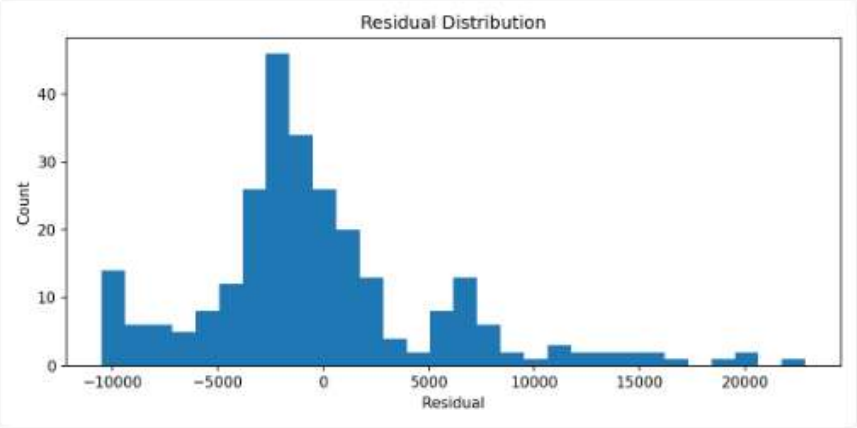
Model loaded successfully.

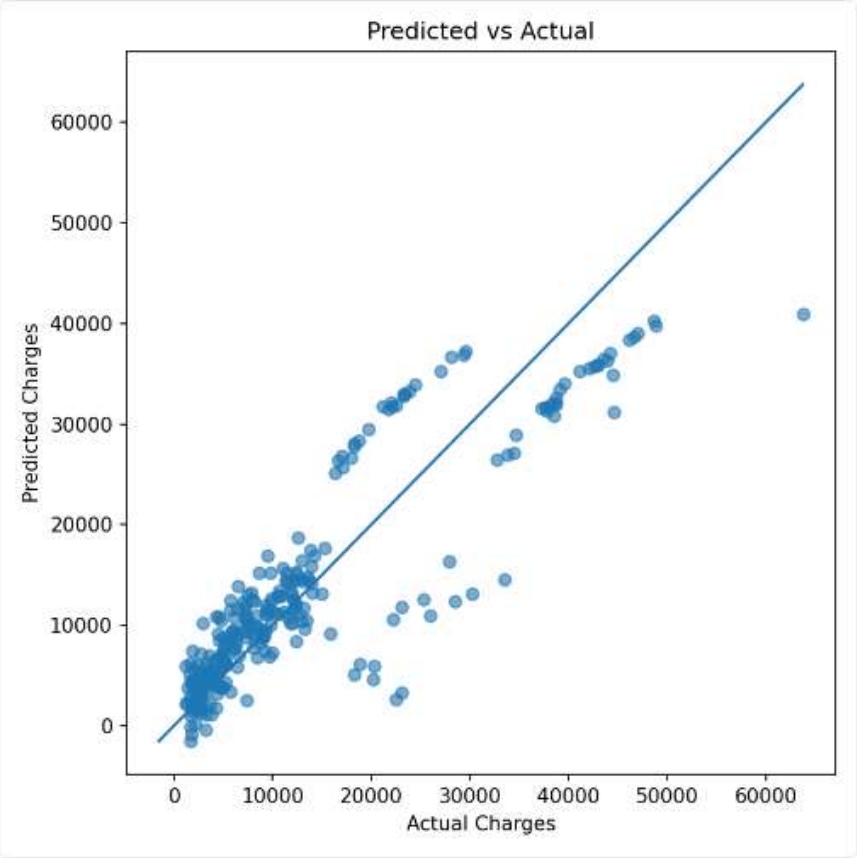
Predicted Charges:

$—

Model Metrics

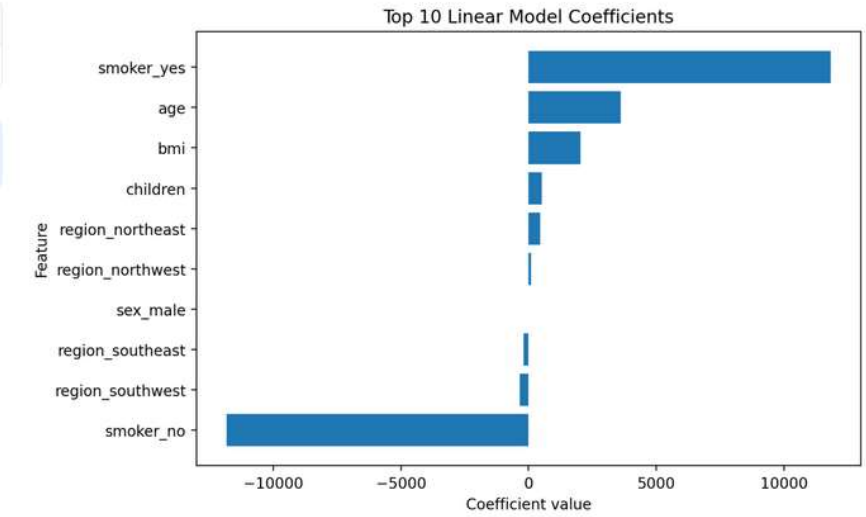
MAE: 4181.19 | RMSE: 5796.28 | R^2: 0.7836



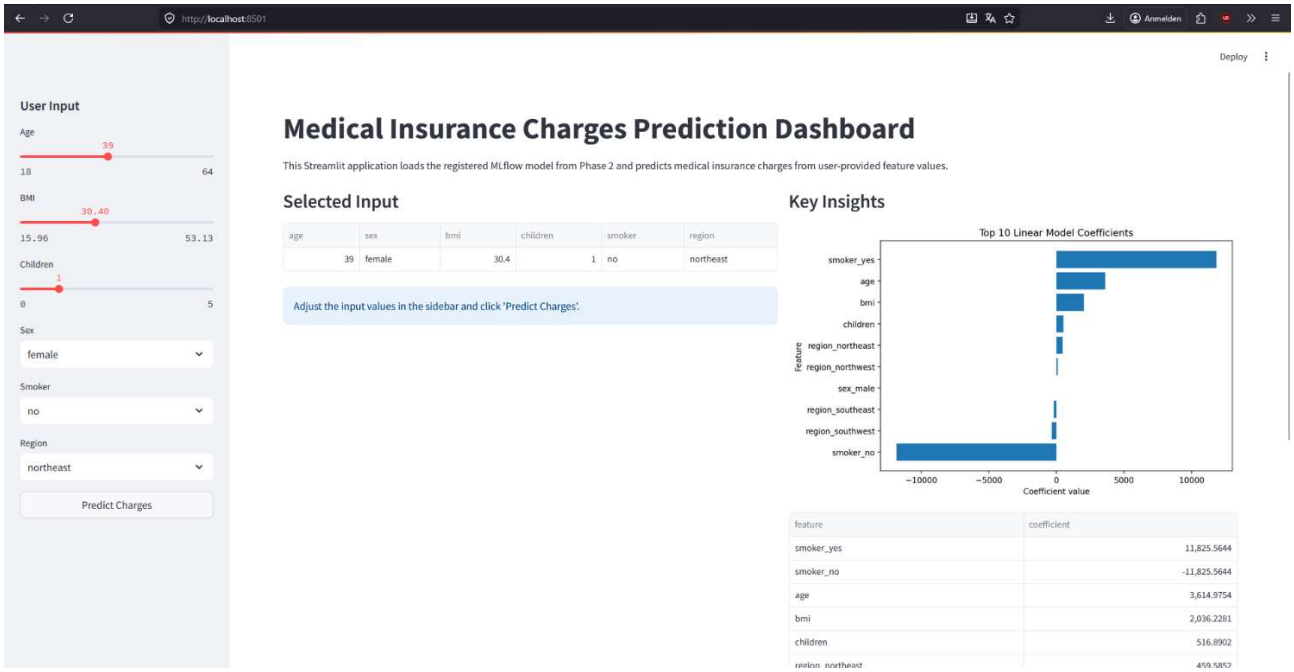


Key-Insights:

Key Insights



feature	coefficient
smoker_yes	11,825.5644
smoker_no	-11,825.5644
age	3,614.9754
bmi	2,036.2281
children	516.8902
region_northeast	459.5852
region_southwest	-350.2141
region_southeast	-198.2791
region_northwest	88.9079
sex_male	-9.2958



Architecture / Loading the Model from MLflow

The application is implemented as a Streamlit dashboard and serves as a user interface for the model. At startup, it connects to the MLflow tracking server via the URI http://127.0.0.1:5000. Afterwards, the registered model **BestRegressor_v1** is loaded. The application first attempts to use the production version. If this is not available, the latest available version is automatically loaded as a fallback.

How Do the Inputs Flow into the Prediction?

The inputs are collected via the Streamlit sidebar and include variables such as age, BMI, number of children, sex, smoker status, and region. These values are converted into a single-row pandas DataFrame. The DataFrame is passed directly to the model.predict() function. The MLflow model already contains the complete preprocessing pipeline, which ensures that categorical variables are automatically encoded and the prediction is computed correctly.

Discussion: What Do the Key Insights Show?

The key insights are based on the coefficients of the linear model. It can be observed that smoking has the strongest impact on costs. Age and BMI also have a clearly positive effect on insurance charges. The number of children has only a minor influence. Region and sex play a comparatively smaller role.

What Would Be a Meaningful Improvement?

A meaningful improvement would be the use of more powerful models such as Random Forest or XGBoost. In addition, uncertainty estimation should be integrated directly into the MLflow model. The use of MLflow could be improved by introducing clear stages and automated model comparisons. For production use, a REST API would be beneficial. Additionally, monitoring for predictions and data drift should be implemented.